

What is Target Encoding?

Target Encoding replaces each category with the mean of the target variable for that category.

◆ Example (for classification):

Category Target Encoded Value

A	1	0.75 (avg of A's targets)
A	0	0.75
B	1	0.33
B	0	0.33

✓ Use When:

- Dealing with high-cardinality categorical variables (many unique categories)
 - Models that can benefit from numerical correlations (e.g., regression, tree-based models)
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✗ Be Careful:

- Can cause data leakage — always use cross-validation or smoothing to avoid overfitting.
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🔧 Tools:

- `category_encoders` library → `TargetEncoder`

- Custom implementation with groupby & mean